Gait Type Classification Using Smart Insole Sensors

1st Sang-Il Choi
Dept. Computer Science and Engineering (of Dankook University)
Yongin-si, South Korea
choisi@dankook.ac.kr

2nd Sung-Sin Lee
Dept. Data Science (of Dankook University)
Yongin-si, South Korea
leesungsin@gmail.com

3rd Hee-Chan Park
Dept. Applied Computer Engineering (of Dankook University)
Yongin-si, South Korea
hchan11@naver.com

4th Hyunil Kim
Dept. Software Science (of Dankook University)
Yongin-si, South Korea
gusdlf93@naver.com

***Abstract*—In this paper, we propose a gait type classification method using various sensors in a smart insole. The measured data are normalized to the unit step of the same length in order to reduce the variation of the speed according to the measurement point and the situation even within the same gait type. From the normalized data of the individual sensors, the discriminant features useful for gait type classification are extracted by using the Null-Space Linear Discriminant Analysis (NLDA), one of the representative discriminant analysis methods. As a result of experiments on the data measured for the seven gait types, our method gives a good performance of gait type classification.**

***Index Terms*—gait type, classification, discriminant analysis, smart insole, sensors**

I. Introduction

Gait is one of the representative behaviors of human beings, and analysis of gait pattern can be applied in many applications such as biometrics, rehabilitation medicine or health care. As one of the fields related to gait analysis, gait type classification is being studied for the diagnosis of Parkinson's disease [1], sports analysis [2], or the walking aids for the elderly [3] based on gait data acquired using sensors. Gait types have characteristics that are easily affected by factors such as physical differences due to physical characteristics or differences in speed and terrain. These characteristics make it difficult to classify gait patterns because it increases the variation within the same gait type.

The gait type classification system consists of a sensor module for acquiring data and an application module for classifying the gait type based on the acquired data. Sensors used in gait classification are mainly video sensors [4], electromyographic sensors [5], plantar pressure sensors [6], accelerometers [7], [8], gyro sensors [9], [10], etc. However, most sensors have limitations in measuring walking data only in a restricted environment due to sensor size, installation inconvenience, and so on [6]. Recently, the development of the wearable sensor technology has led to the weight reduction and simplification of the equipment to be used for the measurement of the gait data. Therefore, researches on gait type classification that is less constrained by place and behavior are being actively carried out. In particular, gait type classification methods using smart phones, smart watch acceleration sensors, and gyro sensors have been performed [8], [11]. In recent years, the smart insole with pressure sensor, acceleration sensor and gyro sensor has been developed and used for walking type classification research [12].

In this paper, we propose a gait classification method that classifies the type of gait data in which various variations exist by using sensors mounted on smart insole. As a smart insole for obtaining gait data, FootLogger, we used a commercial smart insole developed by 3L-Labs Co., Ltd. (Seoul Korea). The FootLogger is equipped with a pressure sensor, acceleration sensor and gyro sensor in one insole. The data acquired from each sensor during walking is transmitted to the smart phone via Bluetooth.

For the gait type classification, we preprocessed to transform the data measured in the sensor into a form suitable for the discriminant analysis and the features useful for the gait type classification are extracted using a discriminant analysis method [13]. In [12], the gait type was effectively classified using only the smart insole pressure sensor. In this paper, we perform gait type classification experiments using acceleration sensor and gyro sensor data in addition to pressure sensor, and investigate the possibility of developing a more effective gait classification method by comparing classification performance according to sensor type.

This paper is organized as follows. Section 2 presents the data acquisition using a smart insole and the data preprocessing. Section 3 explains how to extract features for classification from data. In Section 4, we describe experimental results of gait type classification for each sensor data and conclusions are followed in Section 5.

II. Data Acquisition and Pre-processing

In this paper, 'FootLogger' (Fig. 1), which is a commercial smart insole used, has 8 pressure sensors, 3-axis accelera-

This research was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. 2018R1A2B6001400) and was also supported by the MSIT(Ministry of Science and ICT), Korea, under the National Program for Excellence in SW supervised by the IITP(Institute for Information and communications Technology Promotion) (2017-0-00091).

Authorized licensed use limited to: Malta College of ArtsScience&Tech. Downloaded on November 24,2025 at 11:34:55 UTC from IEEE Xplore. Restrictions apply.

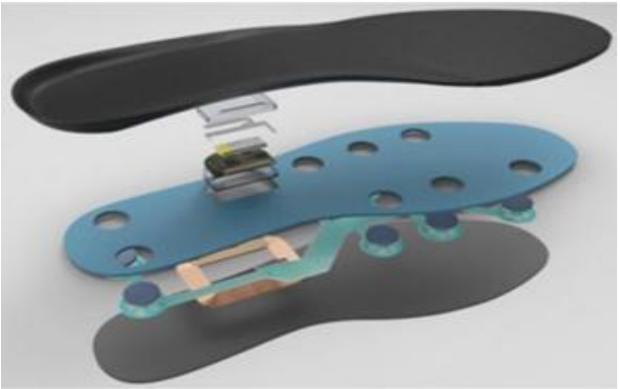

Fig. 1. **Sensor structure of the smart insole 'FootLogger'.**

Fig. 2. **Gait cycle of seven steps.**

tion sensor and 3-axis gyro sensor [12]. The insole sensors mounted on both shoes measure the data at a sampling rate of 100Hz. In the case of a pressure sensor, the value is measured as 0, 1, 2 depending on the intensity of the pressure, where 0 is the pressure-free state, called the swing phase, and 1 and 2 represent the stance phase, i.e., the pressure at which the foot is touching the ground.

As shown in Fig. 2, the gait cycle is generally divided into seven stages: 'heel strike', 'foot flat', 'mid stance', 'heel off', 'toe off', 'mid swing', and 'late swing' [14]. The 'heel strike' is the start of the gait cycle. The heel is in contact with the ground. The 'foot flat' is the state in which the entire sole touches the floor. The 'mid stance' is the state until the center of the body moves to the forefoot. With the 'heel off' and 'toe off' falling from the ground, 'mid swing' and 'late swing' all float in the air and one gait cycle ends in late swing. Alternatively, one gait cycle can be divided into a swing phase with one foot floating and a stance phase with the ground attached.

Figure 3 shows an example of gait data measured using 'FootLogger'. In Fig. 3, the gait cycle can be confirmed by alternating the swing phase in the left foot and the right foot based on the pressure sensor data. In this paper, we define a step from the start point of the swing phase to the end point of the stance phase based on the left foot as in [12]. The gyro sensor and the acceleration sensor measure sensor values for three directions x, y, and z in the insole of each foot.

	Date	Pressure Sensors																Measured: Acceleration & Gyroscope			
		Left								Right								Left		Right	
Stance Phase	17:59,11.962	2	1	1	0	0	1	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,11,972	2	1	0	0	0	1	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,11,982	1	0	0	0	0	1	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,11,992	1	0	0	0	0	1	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,12,002	0	0	0	0	0	1	0	0	0	0	0	2	0	0	1	2	.	.	.	.
Swing Phase	17:59,12,012	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,12,022	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,12,032	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,12,042	0	0	0	0	0	0	0	0	0	0	2	2	0	0	1	2	.	.	.	.
	17:59,12,052	0	0	0	0	0	0	0	0	1	2	1	2	0	0	1	2	.	.	.	.
	17:59,12,062	0	0	0	0	0	0	0	0	1	2	1	2	0	0	1	2	.	.	.	.
	...																				
Stance Phase	17:59,12,532	0	0	0	2	0	0	1	2	2	2	0	0	0	0	0	0	.	.	.	.
	17:59,12,542	0	0	0	2	0	1	2	2	2	2	0	0	0	0	0	0	.	.	.	.
	17:59,12,552	0	0	0	2	0	1	2	2	1	2	0	0	0	0	0	0	.	.	.	.
	17:59,12,562	1	1	1	2	1	1	2	2	1	2	0	0	0	0	0	0	.	.	.	.
	17:59,12,572	2	2	1	2	2	1	2	2	1	1	0	0	0	0	0	0	.	.	.	.
	17:59,12,582	2	2	1	2	2	2	2	2	0	0	0	0	0	0	0	0	.	.	.	.
	...																				
	17:59,12,742	2	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	.	.	.	.
	17:59,12,752	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.	.	.	.
	17:59,12,762	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.	.	.	.
	17:59,12,772	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.	.	.	.
Swing Phase	17:59,12,782	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,12,792	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	2	.	.	.	.
	17:59,12,742	0	0	0	0	0	0	0	0	1	2	1	2	0	0	1	2	.	.	.	.

1 Sample

Fig. 3. **An example of gait data measured using 'FootLogger'.**

On the other hand, even if the same person is walking, the speed of walking may vary during data measurement or when data is attempted to be measured. This change in walking speed interferes with the extraction of gait characteristics to distinguish the type of gait. In order to extract features that are less sensitive to the situation, the unit steps from every swing phase to the stance phase are resized based on the time (t) of the shortest unit step (in this paper, $t = 63$), and normalization is performed so that the lengths for all the steps are the same. As a result, the pressure sensor data was stored in an array of 63×16, and the acceleration sensor and the gyro sensor were stored in an array of 63×6, respectively. Then, the measured value of each sensor of the normalized unit step was stored as a vector of 1008x1 (pressure sensor) or 378x1 (acceleration, gyro sensor) using a lexicographic ordering operator. The normalized measurements of each sensor for each step were then stored as a vector of 1008×1 (pressure sensor) and 378×1 (acceleration and gyro sensor) using a lexicographic ordering operator.

The data were collected by walking or running on flat, stair, and hill to get a large set of terrain variation and speed variation. The data to be used in the experiment consisted of 7 types of gait types from 13 people: walking on the flat ground (WK), fast walking (FWK), running (RUN), hill climbing (RA) and descending (RD), stair climbing (SA) and stair descending (SD). The data were collected at a natural speed. In the case of WK, FWK and RUN, the data were collected for 3 minutes. RA and RD were measured for 2 minutes. In the case of SA and SD, data were collected by walking the stairs for 1 minute due to the limitation of the place.

Sensor data may include noise due to factors such as the measurement environment, the potential difference within the

Authorized licensed use limited to: Malta College of ArtsScience&Tech. Downloaded on November 24,2025 at 11:34:55 UTC from IEEE Xplore. Restrictions apply.

TABLE I
INFORMATION ON THE MEASURED DATA SAMPLES [12] [12]

Terrain	Gait type	Number of steps	Measuring time
Flat	walking(WK)	2193	3mins
Hill	uphill(RA)	1493	2mins
Hill	downhill(RD)	1483	2mins
Stairs	climbing stairs(SA)	760	1min
Stairs	step down(SD)	838	1min
Flat	run(RUN)	3154	3mins
Flat	fast walking(FWK)	2530	3mins

module, and the heat of the sensor. If one of the eight pressure sensors exhibits a nonzero measurement due to this noise in the swing phase, this results in a swing phase misdetction. Since the malfunction of this sensor shortens the normal swing phase time, swing phases having a length of less than 0.5 second are regarded as an abnormal state, and unit step samples including such swing phase are excluded from the experiment. As a result, a total of 13,532 steps of data are used in the experiment and the information on the measured data samples is shown in Table 1.

III. FEATURE EXTRACTION FOR CLASSIFICATION

To extract useful features for gait type classification from preprocessed gait data, Nullspace LDA (NLDA) [13], which is effective for high-dimensional data, is used as the discriminant feature extraction method.

First, for we define the between-class scatter matrix S_B and the within-class scatter matrix S_W for N_i samples composed of c_i, $i = 1, 2, .., C$ classes as follows.

$$\begin{aligned} S_B &= 1/N \sum_{i=1}^{c} N_i(\mu_i - \mu)(\mu_i - \mu)^T \\ S_W &= \sum_{i=1}^{c} \sum_{\mathbf{x}_k^c \in \{c_i\}} (\mathbf{x}_k - \mu_i)(\mathbf{x}_k - \mu_i)^T \end{aligned} \tag{1}$$

Here, $N = \sum_{i=1}^{C} N_i$. $\mathbf{x}_k$ is the k-th sample belonging to class c_i, and μ and μ_i are the mean of the entire sample and the mean of the samples belonging to class c_i, repectively. In order to maximize the discrimination power between classes, NLDA projects the samples in null space of S_W, and then obtains projection vectors satisfying the following objective function so that the variance between the means of each class becomes the maximum.

$$W_{NLDA} = \arg \max_{|W^T S_W W|=0} |W^T S_B W| \tag{2}$$

W_{NLDA} is a projection matrix composed of n' projection vectors $\mathbf{w}_t, t = 1, .., n'$, and a feature vector $\mathbf{y}$ for sample $\mathbf{x}$ can be obtained as follows.

$$\mathbf{y}_k = W_{NLDA}^T \mathbf{x}_k \tag{3}$$

Figure 3 shows the whole process of gait type classification using a smart insole sensor.

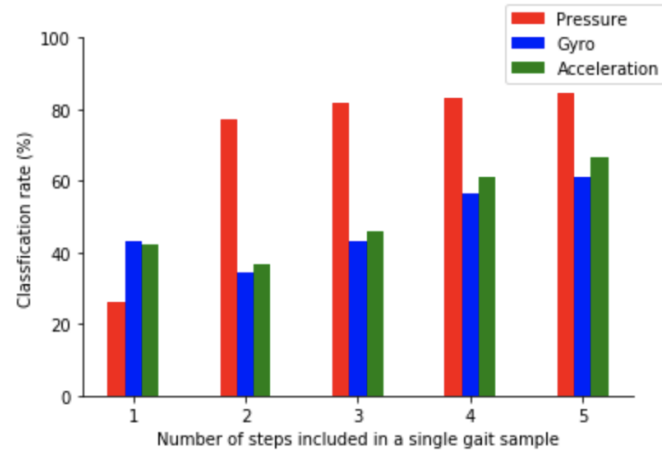

Fig. 4. **Classification performance for different numbers of steps included in single sample.**

IV. EXPERIMENTAL RESULTS

In order to evaluate the performance of gait type classification using a sensor in a smart insole, we preprocessed the measurements of the acceleration sensor, gyro sensor and pressure sensor of the FootLogger and extracted the discriminant features using NLDA. The training data set was composed of 1000 samples randomly selected from the total data samples, and another randomly selected 1000 samples were used as the test set. Prior to extracting NLDA features, all data samples were normalized to have zero-mean and unit standard deviation with the mean and standard deviation of the training data, and a classifier using extracted features NN (nearest neighborhood rule) classifier. The results of this experiment are shown as the average of the classification rates obtained by repeating 20 times.

In order to find out how many steps are needed to extract the gait characteristics for gait classification, we measured the classification rate for each sensor from the gait sample consisting of the sensor measurement for one step to the gait sample including the measurement value for five steps.

Figure 4 shows the classification rates for each case. Comparing each sensor in Fig. 4, the pressure sensor data showed 84.41% higher classification performance when using a sample consisting of five steps, and showed better performance than other sensors. Especially, in the case of the pressure sensor data, classification performance is greatly improved when the feature is extracted from two steps rather than one step. It is considered that a significant gait pattern is found when walking at least two steps. In addition, the larger the number of steps of a sample, the higher the classification performance becomes because the larger the dimension of the sample, the more the null space of S_W is secured in finding the feature space of NLDA.

On the other hand, the acceleration and gyro sensors showed lower classification rates than the pressure sensors, but showed more than 60% classification performance in samples contain-

Authorized licensed use limited to: Malta College of ArtsScience&Tech. Downloaded on November 24,2025 at 11:34:55 UTC from IEEE Xplore. Restrictions apply.

ing five steps. Unlike the pressure sensor data, which have only three kinds of values of 0, 1, and 2, the acceleration sensor and the gyro sensor have a value between -32768 and 32767 and a correlation between measured values is large. Therefore, synergy effects can be expected if data of heterogeneous sensors such as pressure sensor and acceleration sensor, or pressure sensor and gyro sensor are used together.

V. Conclusion

In this paper, we present a method to measure gait data using a smart insole sensor and to classify the gait type by using the extracted features based on discriminant analysis. Experimental results on the data measured for 13 people showed that the information available for gait type classification can be extracted by discrimination analysis of pressure sensor, gyro sensor and acceleration sensor data of the smart insole, and especially, the performance of the pressure sensor data was the highest. In consideration of classification performance of individual sensors and heterogeneity between sensors, it is expected that classification performance will be improved by using different types of gait sensor data together. We plan to conduct research on this in the future.

References

[1] J. Klucken, J. Barth, P. Kugler, J. Schlachetzki, T. Henze, F. Marxreiter, Z. Kohl, R. Steidl, J. Hornegger, B. Eskofier *et al.*, "Unbiased and mobile gait analysis detects motor impairment in parkinson's disease," *PloS one*, vol. 8, no. 2, p. e56956, 2013.

[2] D. Gouwanda and S. Senanayake, "Emerging trends of body-mounted sensors in sports and human gait analysis," in *4th Kuala Lumpur International Conference on Biomedical Engineering 2008*. Springer, 2008, pp. 715–718.

[3] Y. Hayashi and K. Kiguchi, "Stairs-ascending/descending assist for a lower-limb power-assist robot considering zmp," in *Intelligent Robots and Systems (IROS), 2011 IEEE/RSJ International Conference on*. IEEE, 2011, pp. 1755–1760.

[4] M. Heydarzadeh, J. Birjandtalab, M. B. Pouyan, M. Nourani, and S. Ostadabbas, "Gaits analysis using pressure image for subject identification," in *Biomedical & Health Informatics (BHI), 2017 IEEE EMBS International Conference on*. IEEE, 2017, pp. 333–336.

[5] H. Huang, T. A. Kuiken, R. D. Lipschutz *et al.*, "A strategy for identifying locomotion modes using surface electromyography," *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 1, pp. 65–73, 2009.

[6] S.-D. Min and C.-K. Kwon, "Step counts and posture monitoring system using insole type textile capacitive pressure sensor for smart gait analysis," *Journal of The Korea Society of Computer and Information*, vol. 17, no. 8, pp. 107–114, 2012.

[7] S.-Y. Kim and G.-I. Kwon, "Gravity removal and vector rotation algorithm for step counting using a 3-axis mems accelerometer," *Journal of the Korea Society of Computer and Information*, vol. 19, no. 5, pp. 43–52, 2014.

[8] W. Wu, S. Dasgupta, E. E. Ramirez, C. Peterson, and G. J. Norman, "Classification accuracies of physical activities using smartphone motion sensors," *Journal of medical Internet research*, vol. 14, no. 5, 2012.

[9] T. T. Ngo, Y. Makihara, H. Nagahara, Y. Mukaigawa, and Y. Yagi, "Similar gait action recognition using an inertial sensor," *Pattern Recognition*, vol. 48, no. 4, pp. 1289–1301, 2015.

[10] T. Zhang and G. Venture, "Individual recognition from gait using feature value method," *Cybernetics and Information Technologies*, vol. 12, no. 3, pp. 86–95, 2012.

[11] H. Sun and T. Yuao, "Curve aligning approach for gait authentication based on a wearable accelerometer," *Physiological measurement*, vol. 33, no. 6, p. 1111, 2012.

[12] W.-D. Seo, S.-S. Lee, W.-Y. Shin, and S.-I. Choi, "Gait type classification using pressure sensor of smart insole," *Journal of The Korea Society of Computer and Information*, vol. 23, no. 2, pp. 17–26, 2018.

[13] H. Cevikalp, M. Neamtu, M. Wilkes, and A. Barkana, "Discriminative common vectors for face recognition," *IEEE Transactions on pattern analysis and machine intelligence*, vol. 27, no. 1, pp. 4–13, 2005.

[14] S. Shultz, P. Houglum, and D. Perrin, *Examination of Musculoskeletal Injuries with Web Resource*. Human Kinetics, 2015.

Authorized licensed use limited to: Malta College of ArtsScience&Tech. Downloaded on November 24,2025 at 11:34:55 UTC from IEEE Xplore. Restrictions apply.